

Mohammad Taiyab Khan

Data & AI Engineer | Co-Founder, Autostrata.ai

I build AI products that move from idea → demo → deployed.

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Portfolio: autostrata.ai • Egham, Surrey, UK • Graduate Visa to Jan 2027 • Open to UK Skilled Worker sponsorship & international relocation

PROFESSIONAL SUMMARY

Data & AI Engineer with an MSc in Data Science & Analytics (Merit) from Royal Holloway, University of London, and 3 years of combined experience across data analytics, machine learning, and full-stack engineering. I take ideas from concept to production: a recommendation system that lifted user engagement by 50% at Gurucool; a face-verified attendance platform that cut payroll errors by 72% across 45 employees; a voice-activated emergency app with 96.8% wake-word accuracy. Comfortable across Python, SQL, Power BI, R, classical ML, and modern web stacks (React, Next.js, Node.js, FastAPI). Equally happy presenting findings to non-technical stakeholders.

PROFESSIONAL EXPERIENCE

Co-Founder & Data/AI Engineer | *Autostrata.ai*

May 2025 – Present • Egham, UK

- Co-founded a UK-based AI consultancy (registered Oct 2025) helping startups and SMEs ship AI-powered products: agentic AI, automation, data dashboards, and modern web/mobile apps. Repeat clients with extended engagements.
- Clock-in Pro (paid client, ongoing): designed and shipped a face-verified, GPS-geofenced attendance platform — React/TypeScript frontend, Node/Express backend, PostgreSQL with Drizzle ORM, Python microservice with DeepFace and OpenCV. Outcomes after 6 months in production: fake clock-ins down 83% (12 → 2 per month), payroll adjustments down 72% (7.5% → 2.1%), ~4.5 admin hours saved per week. Scaled to 45 employees across 3 sites; 99.6% uptime, p95 API latency 820ms, 98.3% face-verification accuracy.
- MummyHelp (paid client, follow-on): built a voice-activated SOS mobile app — React Native/Expo frontend, Node/Express backend, Supabase (PostgreSQL with Row Level Security), Python FastAPI microservice running SpeechBrain ECAPA-TDNN for speaker verification, Picovoice Porcupine for on-device wake-word detection. 96.8% wake-word accuracy, FAR 1.1% / FRR 2.4%, average SOS delivery 3.5s (61% faster than baseline), 99.3% crash-free sessions. Privacy-first: no continuous cloud streaming.
- HEART-Eco (Eco Heart AI): built a Next.js 14 dashboard for a custom multi-pillar economic scoring model (Housing-Health, Energy-Education, Affordability, Rate, Trade) covering country/year benchmarks and 2026–2030 forecasts. Integrated Recharts visualisations and an OpenAI Assistants 'Ask AI' panel for natural-language model exploration. Produced investor-pitch visuals (HEART pyramid schema) and supporting ensemble ML model.
- Sartorial London (contract): extended a Sharetribe Flex marketplace for outfit rentals — implemented Stripe security-deposit / charge-later flow, fixed 5-day rental booking logic, added end-to-end shipping fee support, custom transaction states (Sent / Received / Returned), and revamped landing page with a curated 'Featured Fits' carousel.
- Additional client work: marketing sites for C&A Solution and Empire Exchange — Lighthouse 98, LCP 1.2s, CLS 0.01; contact-form conversion +200% in 3 weeks. Reception AI prototype: greets visitors, captures details + photo, issues digital visitor cards. Converted an academic paper into a sellable AI solution using LLM + n8n automations and RAG/prompt design.
- Stack: Python (Pandas, NumPy, FastAPI, DeepFace, OpenCV, SpeechBrain), React 18, Next.js 14, TypeScript, Node.js/Express, PostgreSQL, Supabase, Drizzle ORM, Docker, Stripe, Vercel, Railway, Replit, OpenAI Assistants API, n8n.

AI Content Analyst | *Outlier (Contract)*

Jan 2024 – Mar 2024 • London, UK

- Evaluated AI-generated outputs for factual accuracy, domain relevance, and quality across STEM and educational content pipelines — flagging incorrect responses, identifying root causes, and writing structured corrections. Contributed to RLHF-style annotation workflows used to train downstream LLMs.
- Worked primarily in Python; assessed mathematical reasoning, code outputs, and natural-language responses against quality rubrics.

Data Analyst | *Gurucool XYZ Pvt Ltd*

Jan 2022 – Aug 2023 • New Delhi, India

- Led development of a Python-based hybrid recommendation system (collaborative + content-based filtering) for Gurucool's BITS short-form content feature, increasing user engagement by 50% and reducing churn by 40%; co-authored a research paper on the methodology.
- Integrated OpenAI GPT API (Davinci, Babbage, Ada, GPT-3.5 Turbo) with custom prompt engineering to power an in-app Q&A chatbot — improving user satisfaction scores and session depth (built during the early ChatGPT era).
- Built Power BI dashboards (MIS reporting) consolidating cross-team data sources for senior management, enabling weekly business reviews and KPI tracking.

- Performed user-behaviour analysis using Google Analytics, Google Tag Manager, Microsoft Clarity, and Hotjar; identified UX drop-off points and informed UI redesign for the Padhaai app.
- Conducted cross-platform social media analytics across LinkedIn, Instagram, and Facebook using Supermetrics, Dataslayer, and Meta Analytics; created a unified Google Data Studio dashboard combining multi-source performance data.
- Built a Python web-scraper using BeautifulSoup to extract and structure data on 10,000+ Indian colleges, supporting content targeting and personalisation strategy.

Retail Assistant | Primark

Oct 2023 – Present • Camberley, UK

- Part-time alongside studies and freelance work. Cash office operations, stock management, and customer service. Recognised as Employee of the Quarter.

EDUCATION

MSc Data Science & Analytics — Merit

Sep 2023 – Sep 2024

Royal Holloway, University of London

- Modules: Machine Learning, Online Machine Learning, Large-Scale Data Storage & Processing (Spark, HDFS), Data Visualisation & Exploratory Analysis, Databases, Ethical Issues in AI, Advanced Computing.
- Dissertation (supervised by Prof. Daniel O'Keeffe): Predictive Power of Social Media — Predicting EURO 2024 outcomes via Twitter sentiment analysis, comparing classical ML, lexicon-based, and zero-shot LLM approaches. Findings later published as a research paper on Zenodo.

Diploma in Data Science — First Division (7.6 CGPA)

Mar 2021 – Mar 2023

Indian Institute of Technology (IIT) Madras

- Modules: Machine Learning Fundamentals, Business Analytics, Tools for Data Science. Capstone on shopper-behaviour modelling.

BSc Computer Science & Mathematics — First Division (8.28 CGPA)

Sep 2020 – Sep 2023

Jamia Millia Islamia, New Delhi, India

- Modules: Database Management Systems, Data Structures, Probability & Statistics. Final project on Alzheimer's detection from MRI scans (84% accuracy, F1 = 0.8).

SELECTED PROJECTS

AI ENGINEERING & PRODUCT

Clock-in Pro — Face-Verified Attendance Platform — github.com/data-professional-taiyabkhan/clock-in-pro

- Solo full-stack build: React/TS + Tailwind + shadcn/ui frontend, Node/Express (TypeScript) backend, PostgreSQL with Drizzle ORM, Python DeepFace/OpenCV microservice. Deployed via Replit/Railway.
- Live measurable outcomes: 83% reduction in fake clock-ins, 72% reduction in payroll adjustments, 99.6% uptime across 45 staff and 3 sites.

MummyHelp — Voice-Activated Emergency App — github.com/data-professional-taiyabkhan/MummyHelpIA

- Wake-word detection (Picovoice Porcupine, on-device KWS) + speaker verification (SpeechBrain ECAPA-TDNN, cosine-similarity matching, 1.2s verify time) via Python FastAPI.
- Production-grade backend: JWT auth, rate-limiting (100 req/15min), Helmet security headers, Supabase Row Level Security, Docker containerisation, EAS Build for app store distribution.

HEART-Eco — Economic Model Dashboard — heart-eco.vercel.app

- Composite multi-pillar economic scoring framework with 2026–2030 country forecasts. Next.js 14 dashboard with Recharts and OpenAI Assistants 'Ask AI' panel for natural-language exploration. Supporting ensemble ML model predicts HEART score and key parameters.

UK Visa Sponsorship Companies Mapping App — github.com/data-professional-taiyabkhan/UKSponsorshipCompany

- Built to solve a real problem (mine and other international students'): an interactive Leaflet.js map of UK companies holding sponsor licences, with marker clustering and town/city filtering. Flask backend with Google Maps API geocoding, deployed on cloud. 3,674 LinkedIn impressions on launch post.

DATA SCIENCE & MACHINE LEARNING

Predicting EURO 2024 via Social Media Sentiment (MSc Dissertation)

- Built four classifiers — TF-IDF + Logistic Regression (86% accuracy, AUC 0.927), VADER, three-class LR, and a zero-shot Claude Haiku LLM with a custom football-aware prompt. Only the LLM detected the genuine win-day vs loss-day sentiment shift ($t = 2.52$, $p = 0.012$).
- Key insight: 79% of football tweets are factually neutral (score updates, lineup news), which forces binary classifiers to inflate noise and bury real signal. Demonstrated that benchmark accuracy and domain sensitivity measure different things. Published as a peer-reviewed paper on Zenodo (DOI: 10.5281/zenodo.19675434).

E-commerce Shopper Behaviour Prediction (IIT Madras Capstone)

- Trained ensemble of XGBoost, AdaBoost, Logistic Regression, K-Means, and Decision Trees to predict purchase intent on real transaction data — 81% accuracy after handling missing values, class imbalance, and feature engineering.

Alzheimer's Detection from Brain Scans (BSc Final Project)

- ML classifier with computer vision techniques on MRI imagery; 84% accuracy and F1 = 0.8. Parallel statistical analysis on Alzheimer's Disease and Healthy Aging Data.

Forest Fire Risk Prediction System

- End-to-end ML pipeline on the Algerian Forest Fire dataset — EDA, preprocessing, model training (Logistic Regression, Random Forest, Gradient Boosting), Flask web app deployed on Render for real-time predictions.

DATA ANALYTICS & VISUALISATION

NBA Shot Selection Shift Analysis (1996–2025)

- Season-level dataset using Python (pandas, nba_api) covering FGA/FGM, 3PA/3PM, FG% by distance bin; normalised for shortened seasons.
- Key finding: 24+ ft attempt share rose from 10.6% → 41.9% (+31.2pp) over 29 seasons; 16–24 ft share fell 25.0% → 4.8% (–20.2pp). Publication-ready visuals (Matplotlib + Canva) for LinkedIn storytelling.

AAPL Multi-Horizon Stock Return & Risk Analysis

- 10-year monthly OHLCV dataset; engineered 1-month, 3-month, 6-month, and yearly return series; quantified risk via volatility and confidence intervals — separating return volatility from estimator uncertainty in a public write-up.

Cyclistic Bike-Share Analysis (Google Data Analytics Capstone)

- Behavioural analysis of 5.78M rides comparing casual vs member usage patterns — ride duration, station popularity, bike-type preferences, and hourly patterns; produced strategic recommendations for membership conversion using SQL, Python, and Power BI.

Twitter Database Redesign & Analysis (PostgreSQL)

- Designed normalised PostgreSQL schema; wrote SQL ETL and analytical queries on language distribution, retweeting habits, and trending hashtags.

Enron Email Network Analysis (Spark + Hadoop)

- Python on Apache Spark and HDFS to analyse communication patterns, sender-receiver relationships, and degree distributions in the Enron corpus.

World Bank International Arrivals — Exploratory Analysis (R)

- Comprehensive EDA in R on World Bank tourism data; ggplot2 static visualisations supporting two research questions with statistical evidence.

TECHNICAL SKILLS

Programming	Python (Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn), R (ggplot2, dplyr), SQL, JavaScript / TypeScript
Databases	PostgreSQL, BigQuery, SQL Server, Supabase, Drizzle ORM, MongoDB (basic)
ML & Statistics	Classification, Regression, Clustering, XGBoost, AdaBoost, Random Forest, Logistic Regression, Decision Trees, K-Means, Neural Networks, Statistical Inference, Confidence Intervals, A/B Testing, Time-Series Analysis
Deep Learning	TensorFlow, PyTorch, Keras, HuggingFace Transformers, CNNs, Computer Vision (DeepFace, OpenCV)
AI / LLMs	OpenAI API (GPT-3.5/4, Assistants), Claude (Anthropic), Prompt Engineering, RAG patterns, n8n automations, SpeechBrain ECAPA-TDNN, Picovoice Porcupine (KWS)
BI & Visualisation	Power BI, Tableau, Google Data Studio, Recharts, Matplotlib, Seaborn, ggplot2
Big Data / ETL	PySpark, MLlib, Hadoop HDFS, ETL pipeline design, BeautifulSoup web scraping
Web & APIs	React 18, Next.js 14, Node.js / Express, FastAPI, REST APIs, Tailwind CSS, shadcn/ui, React Native (Expo), Stripe
Cloud & DevOps	AWS (basic), Google Cloud Platform, Vercel, Railway, Docker, Replit, Postman, GitHub Actions
Analytics Tools	Google Analytics, Google Tag Manager, Microsoft Clarity, Hotjar, Supermetrics, Meta Analytics
Office	Advanced Excel (Pivot Tables, VLOOKUP, INDEX/MATCH), Microsoft Office Suite, Jupyter, Google Colab
Soft Skills	Stakeholder communication, data storytelling, client-facing delivery, analytical thinking, attention to detail

CERTIFICATIONS & CONTINUOUS LEARNING

- Google Data Analytics Professional Certificate — Coursera, Feb 2024

- Machine Learning — Stanford University & Coursera (Andrew Ng), May 2023
- Machine Learning for All — University of London, May 2023
- British Airways Data Science Job Simulation — Forage, Feb 2025
- Foundational Certificate in Data Science — IIT Madras

PUBLICATIONS & RESEARCH

Khan, M. T. (2026). Predictive Power of Social Media: A Comparative Study of Classical ML, Lexicon-Based, and LLM Approaches for Football Sentiment Analysis. Zenodo. DOI: 10.5281/zenodo.19675434